Document 3 -- Details of Participation in Other Competitions / Innovation Forums

Aran Technologies | MBC-3 Registration | Max 300 words

Nirmaan Entrepreneurship and Incubation Programme -- IIT Madras

Aran Technologies is currently in the Nirmaan pre-incubation programme (Phase 1) at the Indian Institute of Technology Madras -- one of India's premier technology institutions. The programme follows the Disciplined Entrepreneurship framework, with structured workbooks covering market segmentation, beachhead market selection, total addressable market sizing, end-user profiling, and go-to-market strategy formulation.

Primary market research was conducted directly with defence and paramilitary end-users, including Border Security Force personnel and Indian Army tactical unit profiles. This research validated a significant capability gap in low-cost indigenous persistent ISR and collaborative aerial surveillance within current border protection operations. The validated beachhead market was identified as BSF border battalions, with a vehicle-mounted tethered drone ISR platform as the primary product offering.

The programme produced two outputs of direct relevance to the MBC-3 submission. First, a validated product-market fit analysis and go-to-market roadmap aligned to Atmanirbhar Bharat defence procurement objectives. Second, the foundational ISR Mission Stack -- a fully operational autonomous drone mission system on PX4 SITL, Gazebo Harmonic, and MAVSDK Python, incorporating LiDAR-based MPC obstacle avoidance and a Flask/SocketIO Ground Control Station. This stack, demonstrated as the programme's technical output, has since been extended and verified for MBC-3 Phase I, forming the single-drone core of the five-drone collaborative radar swarm submission.

Aran Technologies' MBC-3 submission directly applies the technical capability, indigenous development methodology, and defence market understanding being developed through active Nirmaan participation to the Indian Air Force's collaborative drone-based surveillance radar requirement.